

When aggregate RMSE rewards blur: evaluation pitfalls for neural mesh-PDE surrogates

Zhe Jia*
Institute for Geophysics
University of Texas at Austin
Austin, USA
zjia@ig.utexas.edu

Abstract

On sparse, localized physical fields, like the propagating front of a dynamic earthquake rupture, where only $\sim 11\%$ of mesh nodes are active at any instant, we show that **aggregate field RMSE rewards blur**: a model that collapses the front to a near-null field (recovering only $\sim 2\text{--}4\%$ of the true active-front energy, with fewer than 1% of field values exceeding the rupture threshold) attains a *lower* field RMSE than a model that actually tracks the front, purely by matching the $\sim 89\%$ quiescent background. This is not a curiosity. It is manufactured, in our own study, a spurious “regime reversal” between two datasets that vanishes under front-sensitive metrics, and it can flip the apparent winner with the mere choice of rollout horizon. The key diagnostic failure: aggregate field RMSE *cannot distinguish blur from accuracy*, our blur model and the published 3M-step GNS sit comparably on field RMSE (0.56 vs. 0.43), yet their active-front RMSE differs $3\times$ (2.75 vs. 0.99). Only the cross-metric test resolves this: a genuinely accurate model (GNS) wins on *both* metric families simultaneously, while a blur model wins only the aggregate. A second, illustrative pitfall: the head-to-head verdict flips as the baseline is trained from 90k toward 3M steps (N=1 architecture and dataset; supporting evidence only). We also supply a cheap, architecture-agnostic **receptive-field diagnostic** for a trained operator’s *effective* support, yielding one non-obvious fact (global attention is effectively nearly local after training) and showing that **support and accuracy are decoupled along orthogonal axes**. Finally, a 2×2 grid (dataset $\times$ capacity) confirms a positive structural finding: local attention’s front-fidelity advantage holds in all four cells, it is not a dataset or capacity artifact. We do *not* beat the GNS ($\sim 2\times$ better on front metrics) and validate every comparison against the authors’ released rollouts, retracting a preliminary “win” from an under-trained baseline. The contribution is an evaluation methodology for mesh-PDE surrogates: front-sensitive metrics, published-checkpoint baselines, and effective-support measurement without mistaking it for accuracy.

1 Introduction

Neural surrogates promise large speedups over physics solvers for spatiotemporal PDEs on meshes. Message passing [Sanchez-Gonzalez et al., 2020, Pfaff et al., 2021] is the dominant architecture; global attention is an appealing, lower-bias alternative. A common intuition, attention lacks the locality prior of PDEs, so message passing should win in low-data local-dynamics regimes, motivated this study. We set out to *measure* locality directly and to test whether it predicts accuracy. The honest answer surprised us on both counts, and the paper is organized around what we actually found rather than what we expected.

We study dynamic earthquake rupture: slip-rate fields on an unstructured fault mesh [Liu et al., 2025], a sharp-front, transport-dominated system in a low-data regime (10–140 trajectories), plus four synthetic 1D PDEs as controls. We do *not* propose a new state-of-the-art model. Contributions:

- A primary **evaluation pitfall**: aggregate field RMSE *rewards blur* on mostly-quiescent fields, global attention collapses the front yet attains lower field RMSE, manufacturing a spurious “regime reversal” between datasets that front-sensitive metrics erase. We anchor the claim with the published 3M-step

*Corresponding author.

GNS: **field RMSE cannot distinguish blur from accuracy** (blur model 0.56 vs. GNS 0.43 on the aggregate; $3\times$ apart on active-front RMSE). A second, illustrative pitfall: the head-to-head verdict flips with the baseline’s training budget (N=1 architecture and dataset).

- **Support and accuracy are decoupled along orthogonal axes:** effective support (measured by our diagnostic) tracks sensitivity-locality, while rollout accuracy here is set by front-amplitude fidelity, a distinct axis. The front-fidelity ranking follows mixer mechanism type, not the continuous sensitivity-locality scalar (the effectively-local global model is worst, not middling).
- A cheap, architecture-agnostic **receptive-field diagnostic** for the *effective* (post-training) support of a learned PDE operator, robust to its own hyperparameters. It reveals the non-obvious fact that a trained global-attention model is effectively nearly local.
- An **honest, validated** head-to-head against the *published* 3M-step GNS and a 2×2 seed-replicated grid showing local attention’s front-fidelity edge is a structural mixer effect, not a dataset or capacity artifact.

2 Related work

Mesh/graph surrogates. GNS [Sanchez-Gonzalez et al., 2020] and MeshGraphNets [Pfaff et al., 2021] encode locality and relative geometry via message passing; Brandstetter et al. [2022] add pushforward rollout training, and Liu et al. [2025] apply GNS to rupture. **Attention/operators for PDEs.** Softmax attention is ill-conditioned as a learned integral operator; Galerkin/Fourier attention repairs it [Cao, 2021]; physics-aware tokenization (Transolver, Wu et al., 2024) and neural operators [Li et al., 2021] are competitive at scale, consistent with our finding that a global spectral operator does well on smooth 1D PDEs. **Inductive bias.** Low-bias models need more data (ViT, Dosovitskiy et al., 2021); locality/equivariance are the symmetries of local PDEs [Bronstein et al., 2021]. Networks exhibit spectral bias [Rahaman et al., 2019]; GNNs suffer over-squashing [Alon and Yahav, 2021]. **Rollout stability and evaluation.** PDE-Refiner [Lippe et al., 2023] and dissipative operators [Li et al., 2022] target long-horizon accuracy; we use a dissipativity term but focus on measurement and evaluation, where the choice of metric (aggregate vs. front-sensitive) is known to change conclusions for sparse/localized fields.

3 Method

Effective-support diagnostic. For a trained one-step operator F on a mesh with node positions p_i , we perturb the input state at node j and measure the one-step output response at node i ,

$$S(i, j) = \frac{\|F(x + \epsilon e_j)_i - F(x)_i\|_2}{\epsilon},$$

studied vs. distance $d(i, j) = \|p_i - p_j\|$. We normalize each profile by its peak and summarize non-locality by the *far-field influence*, the distance-binned mean normalized S beyond a threshold (2.5 km on rupture; >5 units on the 1D systems). This measures the operator’s *effective* sensitivity support after training, distinct from its nominal architectural support, and, as we show, distinct from its output amplitude fidelity. The probe is one-step (no rollout), applies to any architecture, and is robust to ϵ and the threshold (§5.2).

Front-sensitive metrics. Because the rupture field is mostly quiescent (near-zero away from the propagating front), aggregate field RMSE is dominated by the quiet region. We therefore also report metrics that probe the front: *active-front RMSE* (RMSE restricted to nodes/steps where ground-truth speed exceeds $\text{THR} = 0.5 \text{ m/s}$), *arrival-time MAE* (first threshold exceedance), and two blur signatures, the *peak-speed ratio* and the *active-energy ratio* $\sum_{\text{active}} \|\hat{v}\| / \sum_{\text{active}} \|v\|$ (values $\ll 1$ indicate a collapsed/blurred front).

Models and mixers. Our surrogate maps node tokens through $L=4$ blocks of spatial mixing + a temporal state-space core [Gu and Dao, 2024], with an activity-gated residual head $v_{t+1} = \sigma(g_\theta) \odot (v_t + \Delta_\theta)$. We ablate only the *spatial mixer*, holding the recipe (input-noise injection [Pfaff et al., 2021], dissipativity term, cosine LR, scale-only normalization) fixed: **global** (all-to-all softmax attention); **local** (attention over each node’s $k=16$ nearest neighbors with a relative-position bias on the logits); **message passing (MP)** (an

Table 1: Head-to-head vs. the published 3M-step GNS (full-trajectory rollout, physical units; lower is better). We do not beat it, in- or out-of-distribution. *Ours* reports 3-seed means for M1 and M2 field RMSE and M1 active-front RMSE (same protocol as Table 2); M1 arrival-time MAE matches §5.3. M3 OOD and M1 final-slip are single-seed evaluations (no Table 2 counterpart). GNS wins all metrics on all datasets.

	Ours (local attn)	GNS (3M)
M1 test field RMSE (m/s)	1.137	0.43
M2 test field RMSE (m/s)	0.918	0.40
M3 test field RMSE (m/s), OOD	0.75	0.65
M1 arrival-time MAE (ms)	284	155
M1 final-slip rel- L_2	1.39	0.93
M1 active-front RMSE (m/s)	1.988	0.99

interaction network over the same $k=16$ neighborhood: edge MLP on $[x_i, x_j, p_i - p_j]$, mean aggregation, node MLP). The MP mixer is shallow (one message step per block) and is *not* a proxy for the deep 10–15-hop published GNS.

4 Experimental setup

Data. Dynamic earthquake rupture simulations (EQdyna ground truth) [Liu et al., 2025], ~ 4.7 k mesh nodes, $T=827$ steps (~ 755 active): **M1** homogeneous stress (10/3/6 train/val/test), **M2** stress asperities (30/10/10), **M3** fractal stress (140/10/15, treated as OOD for M2-trained models). Four 1D PDE controls (advection, Burgers, KS, heat) generated spectrally (App. A). **Training and seeds.** To disentangle dataset from capacity, we use a 2×2 grid of $\{\text{M1, M2}\} \times \{\text{large, small}\}$ configs: *large* = $d=128$, depth 4, 6000 steps; *small* = $d=96$, depth 3, 3000 steps; both window 48, batch 2, AdamW 3×10^{-4} cosine, noise 0.02, $k=16$, dissipativity weight 0.3. Each cell is run with multiple seeds (3–5). Full hyperparameters/compute in App. C. **Evaluation.** Autoregressive rollout from the initial frame. We report both aggregate field RMSE and the front-sensitive metrics above, and flag which in each table; we lead with front-sensitive metrics for the reasons in §5.3. We report **mean $\pm$ std over seeds** and **paired same-seed deltas**. **Fair comparison.** We verified our rollout of the released GNS reproduces the authors’ stored rollouts (M1 $\max |\Delta| = 0.009$; M3 fractal 0.566/0.581/1.022 vs. their 0.566/0.584/1.020). A preliminary “win” disappeared once we replaced an under-trained (90k-step) GNS reproduction with the published 3M-step model, quantified in §5.6.

5 Results

5.1 The published GNS wins; we do not beat it

The authors’ 3M-step GNS is $\sim 2\times$ better across all metric families, field RMSE, active-front RMSE, and arrival time, in- and out-of-distribution (Table 1). Training our model far longer (40k steps, $\sim 7\times$) plateaus with no trend toward the GNS, so the gap is not a training-budget artifact. We report this as a clean negative result and spend the rest of the paper on measurement and evaluation.

We present a qualitative illustration of these rollout behaviors in Figure 1. Ground-truth rupture propagates as a sharp, well-defined slip-rate front across the fault (left column). Our local-attention model (middle column), trained with a dissipativity regularizer, is stable and does not blow up, but it suffers from high numerical dissipation and eventually dies out completely by step 256. In contrast, the GNS model (right column, shown here for the 80k-step checkpoint used during our initial experiments) successfully propagates the front over short horizons, but exhibits catastrophic accumulation of high-frequency noise and diverges (blows up) at the long horizon. This visually reinforces the distinction between our stable but overly dissipative surrogate and the GNS baseline’s propensity for long-horizon divergence, which we discuss further in the context of our evaluation pitfalls.

5.2 The diagnostic: global attention is effectively (nearly) local

Over 18 trajectory $\times$ frame rupture samples, far-field influence (>2.5 km, peak normalized to 1) is $3.7 \times 10^{-4} \pm 1.6 \times 10^{-4}$ (global attn), 2.1×10^{-5} (GNS), and $\approx 4 \times 10^{-11}$ (local attn). The headline is not that global attention “leaks domain-wide”, 10^{-4} against a unit peak is a *small* residual floor. Rather, the trained global model has *effectively learned locality on its own*; softmax simply cannot emit exact zeros, leaving a floor that local attention (hard neighborhood) and message passing avoid by construction. The ~ 10 – $18\times$ global/GNS ratio is robust to $\epsilon \in [3 \times 10^{-3}, 10^{-1}]$ and thresholds 2.5–8 km (Fig. 2). As a sanity check that the diagnostic reports known support, on four 1D PDEs it recovers the by-design contrast between a global operator (FNO, far-field 8.7×10^{-4} – 4.7×10^{-3}) and a local one (CNN, exactly 0; Fig. 4, left), architectural facts, included only to validate the probe. The interesting case is rupture, where the diagnostic reveals *effective* support that nominal architecture would not.

5.3 Field RMSE rewards blur; the “regime reversal” is an artifact

The field is sparse. On M1 test, only 11.0% of mesh nodes are active (speed > 0.5 m/s) at a typical timestep, the rupture front is a thin propagating band, so $\sim 89\%$ of the field is near-zero background at any instant. On such a field, aggregate field RMSE is dominated by the quiescent region: a model that matches the background and flatlines the front pays error only on $\sim 11\%$ of the domain. This is the mechanism behind everything below. Concretely, the 89% quiescent background contributes nearly zero to aggregate RMSE for both a blur model and an accurate model (truth $\approx$ prediction ≈ 0 on inactive nodes); all error accumulates in the 11% active region, which is then geometrically diluted in the aggregate. This explains an apparent paradox: the global model recovers only 2–4% of the true active-front energy (energy ratio 0.02–0.04, Table 2), yet attains lower full-trajectory field RMSE than a model that faithfully tracks the front. Aggregate RMSE measures an 11% signal diluted by 89% silence; a model that collapses the front to near-zero “wins” it by doing nothing useful on the part of the field that matters.

Blur wins field RMSE. Our initial reading, from aggregate field RMSE on a single evaluation trajectory, was a clean regime reversal: local beats global on M1 but *loses* on M2. Under a uniform multi-trajectory eval and front-sensitive metrics it dissolves (Table 2). Global attention collapses to a near-null, blurred field, it recovers only ~ 2 – 4% of the true active-front energy (energy ratio 0.02–0.04), and fewer than **1% of its predicted (step, node) pairs exceed the 0.5 m/s rupture threshold** (0.6–0.7%, vs. 18.4% in the ground truth): physically, it predicts almost no rupture, ever, yet attains equal or lower aggregate field RMSE (M2 256-step: essentially tied, 1.63 vs. 1.63; full-trajectory: global lower on *both* datasets, 0.72 vs. 0.92 on M2 and 0.56 vs. 1.14 on M1). On the front itself it is far worse (active-front RMSE: M2 global 2.95 vs. local 2.41; M1 global 2.75 vs. local 1.99; local’s arrival-time MAE is 284 ms on M1 and 706 ms on M2, whereas global’s is degenerate, its front never really arrives). **Under front-sensitive metrics, local attention is best on both datasets, there is no reversal.** That global’s field RMSE on M2 barely moves across data budgets (near-zero variance ± 0.002 , App. B) is a further red flag: a model whose error is independent of the training data is not learning the front, it is coasting on blur.

We visualize the impact of this metric distortion on data efficiency curves in Figure 3. Under the homogeneous M1 dataset (left), the locality prior provides a consistent $\sim 14\%$ rollout RMSE advantage across all trajectory budgets. However, under the more diverse M2 dataset (right), using the aggregate rollout RMSE metric suggests a spurious regime reversal: global attention appears to significantly outperform local attention and exhibit near-zero variance, while local attention appears worse and highly variable. As demonstrated in Table 2, this is a pure artifact of the blur pitfall: global attention achieves its flat, low aggregate RMSE by completely flatlining the rupture front (predicting zero velocity almost everywhere), while local attention is penalized on the aggregate metric for attempting to track the front’s propagation, despite being far superior on front-specific metrics (active-front RMSE of 2.41 vs. 2.95).

The field-RMSE verdict is not even self-consistent across horizons. On M1, global attention *loses* the 256-step field RMSE but *wins* the full-trajectory field RMSE (0.556 vs. local 1.137), because the longer quiescent tail rewards blur even more (Table 2). The front-sensitive ordering, by contrast, is stable across both horizons and both datasets (local $<$ MP $<$ global). A metric whose winner flips with the rollout horizon cannot be the one we judge accuracy by.

Why front-sensitive metrics, and not just “a different ranking”: the GNS anchor. A skeptic could object: if the downstream task were a domain-averaged quantity, global’s lower field RMSE might

Table 2: Spatial-mixer ablation, uniform full eval over *all* test trajectories (M1: 6 traj $\times$ 3 seeds, large config; M2: 10 traj $\times$ 5 seeds, small config; identical recipe). Aggregate field RMSE, especially full-trajectory, favors the *blurred* global model on both datasets, and even flips its own verdict between the 256-step and full-trajectory horizons on M1. Front-sensitive metrics consistently favor local. Energy ratio $\ll 1$ = collapsed front. Lower is better except energy (closer to 1 better); **bold** marks the honest, front-sensitive winner. Arrival-time MAE is degenerate for global (it rarely crosses 0.5 m/s) and is reported in text for local.

dataset	mixer	field RMSE (full)	field RMSE (256)	active-front RMSE	energy ratio
M1	global	0.556	1.342	2.747	0.04
M1	local	1.137	1.259	1.988	0.54
M1	MP	0.917	1.384	2.434	0.20
M2	global	0.720	1.625	2.947	0.02
M2	local	0.918	1.626	2.407	0.29

even be preferable, on what grounds are front metrics the right ones? We close this gap with a consistency test using an independent, published, strong model. The critical observation is that global’s full-trajectory field RMSE (0.556) is *comparable* to the 3M-step GNS’s (0.43), while their active-front RMSE differs $3\times$ (global 2.75 vs. GNS 0.99) and global’s rupture front essentially never materializes ($<1\%$ of field values exceed threshold vs. 18.4% in ground truth). **Field RMSE alone cannot distinguish blur from accuracy:** both the blur model and the genuinely accurate GNS sit near 0.5 m/s on the aggregate, but for entirely different reasons. The distinguishing test is cross-metric consistency: the 3M-step GNS wins on *both* families simultaneously, field RMSE *and* active-front RMSE *and* arrival time (Table 1). A genuinely accurate operator is consistent; the blur model wins one family while failing the other. The anchor argument is therefore not that “global barely misses GNS on field RMSE”, it is that only cross-metric consistency identifies which model is actually accurate, and **front-sensitive metrics are the only way to perform that test.**

5.4 Support does not predict accuracy

Putting §5.2 and §5.3 together: the diagnostic says the trained global model is effectively nearly local, yet it is the *worst* rollout model (it collapses the front). MP is local by construction and is intermediate (active-front RMSE 2.43, energy 0.20). Local attention is most local and best on the front. So effective support and rollout accuracy are governed by different things. We state this mechanistically rather than as an absence of evidence: **support and accuracy are decoupled along orthogonal axes**, the diagnostic measures *sensitivity-locality*, while rollout accuracy here is set by *front-amplitude fidelity*, a distinct property the probe does not observe (the effectively-local global model has near-local sensitivity yet near-zero front amplitude). One might object: the mixer ranking (local $<$ MP $<$ global on active-front RMSE) superficially resembles “more local support = better accuracy.” It does not. **This ranking tracks the mixer mechanism type, not the continuous sensitivity-locality scalar:** the decisive evidence is that the *effectively-local* global variant, whose trained sensitivity is nearly as compact as local attention’s, still ranks worst. If the sensitivity-locality scalar predicted accuracy, the effectively-local global model should rank near local attention; instead it is worst by a wide margin. The front-fidelity ordering thus follows hard-neighborhood relative-position attention vs. mean-aggregation message passing vs. all-pairs softmax attention, a mixer-architecture axis orthogonal to the continuous sensitivity measure. The 2×2 grid (§5.5) confirms this mixer-type effect holds across all four dataset $\times$ capacity cells.

5.5 The 2×2 grid: dataset vs. capacity

Because our M1 and M2 ablations differed in both dataset and config, we ran the full $\{\text{M1}, \text{M2}\} \times \{\text{large}, \text{small}\}$ grid (3 seeds per cell) and report active-front RMSE using the same full-length multi-trajectory eval as Table 2 (Table 3). **Local wins in all four cells.** The advantage is consistent across both datasets and both capacity levels ($\Delta = 0.45\text{--}0.89$), ruling out a pure dataset effect (both M1 cells favor local) and a pure capacity effect (local outperforms global at both the small and large capacity levels). We therefore

Table 3: 2×2 grid, active-front RMSE (front-sensitive; mean over all test trajectories and seeds; lower better). All values use the full-length multi-trajectory eval (826 steps, all test trajectories; same protocol as Table 2). Local wins every cell, ruling out a pure dataset or capacity explanation. Field RMSE still reverses on M1-small (global 0.56 vs. local 1.22), yet active-front RMSE still strongly favors local, the blur pitfall operating within the grid.

	M1-large	M1-small	M2-large	M2-small
global attn	2.747	2.773	2.940	2.947
local attn	1.988	1.888	2.487	2.407

conclude that **the local front-fidelity edge is neither a dataset artifact nor a capacity artifact**, it is a within-recipe structural effect of the spatial mixer.

A secondary observation reinforces the blur argument. On M1-small, aggregate full-trajectory field RMSE reverses: global 0.561 vs. local 1.223 (global appears to win by $2\times$), yet active-front RMSE strongly favors local (1.888 vs. 2.773) and global’s energy ratio is 0.029, near-complete front collapse, indistinguishable from M1-large’s 0.040. Crucially, global’s field RMSE on M1-small (0.561) is essentially identical to M1-large (0.556) and M2-large (0.732 vs. M2-small 0.720): the blur model’s aggregate error is *flat across both capacity and dataset*, confirming it is not learning the front in any configuration, merely coasting on the quiescent background. The blur pitfall thus instantiates itself within the grid: aggregate RMSE produces a false winner both across datasets and across model sizes.

5.6 Evaluation pitfall: baseline-budget sensitivity

Our retracted preliminary “win” came entirely from comparing against a 90k-step GNS; the verdict flips as the baseline is trained from 15k toward 3M steps (Fig. 4). This is an architecture-independent failure mode of surrogate evaluation and motivates validating against *published* checkpoints and released rollouts. It is illustrative (one architecture, one dataset) and we keep it in a supporting role.

6 Discussion

The receptive-field diagnostic is a cheap, robust way to read an operator’s *effective* support, and it produced one genuinely non-obvious fact: a trained global-attention model on rupture is effectively nearly local. But our central, seed-controlled message is a caution. First, **support is not a proxy for accuracy**: the effectively-local global model is the worst, and accuracy is set by an orthogonal axis (front-amplitude fidelity) the diagnostic does not measure. Second, and more actionable, **aggregate field RMSE is a misleading metric for sparse/localized fields**: it rewards a model that blurs the front to near-zero, and it manufactured a spurious “regime reversal” between our datasets that front-sensitive metrics erase. The practical guidance is therefore about *evaluation*, not architecture: measure effective support if you want to understand an operator, but judge accuracy with front-sensitive metrics, and validate baselines against published checkpoints.

Generality of the blur pitfall. The mechanism behind aggregate-RMSE-rewards-blur is purely geometric and not specific to earthquake rupture. Whenever a physical field has a small active fraction $f \ll 1$ (the structure of interest occupies a minority of the domain), aggregate RMSE is dominated by the $(1-f)$ quiescent majority: a model that matches the background and flatlines the structure pays error only on the f active fraction, and that contribution is geometrically diluted. The same trap applies to any domain with sparse or localized structure: turbulent velocity fields (vortex cores occupy a small spatial fraction at any instant), precipitation or wind extremes in climate modeling (extreme events are rare in space-time), wildfire fronts, or any system where the scientifically interesting behavior lives in a spatial or spatio-temporal minority of the full field. In each case, aggregate RMSE rewards a “predict-background” strategy. The diagnostic, check that your metric-winner actually exceeds task-relevant thresholds, and verify consistency against a well-established reference model, requires no architecture changes and applies immediately. We offer the rupture case as a documented, reproducible instance of this general pitfall. We have *not* empirically

verified the mechanism on turbulence, climate, wildfire, or any other domain listed above; the generalization is conceptual and mechanistic, grounded in the geometry of sparse fields, not in cross-domain experiments.

7 Limitations and conclusion

We study one physical domain family (rupture) plus four synthetic 1D PDEs; we do not propose or beat a SOTA model. The 1D component is a sanity check, not a discovery. The MP mixer is shallow and not a GNS proxy. The baseline-budget pitfall is illustrative (N=1 architecture, N=1 dataset). Within these bounds we contribute: a robust effective-support diagnostic; evidence that support and accuracy are decoupled; a primary, well-evidenced evaluation pitfall (aggregate RMSE rewards blur on sparse fields) and one illustrative second pitfall (baseline-budget sensitivity); and a positive structural finding, local attention’s front-fidelity edge, confirmed by a 2×2 seed-replicated grid to be neither a dataset nor a capacity artifact, alongside an honest validated head-to-head that retracts a preliminary win.

A PDE data generation

Advection ($u_t + cu_x = 0$) and heat ($u_t = \nu u_{xx}$) use exact spectral propagation; Burgers ($u_t + uu_x = \nu u_{xx}$, $\nu=0.025$) uses RK4 with 2/3 dealiasing; KS uses ETDRK4. Periodic domains, $N_x=128$; ~ 512 trajectories per system, 80/20 train/test. Code: `paper/burgers_lab.py`, `paper/multisystem_locality.py`.

B Aggregate field RMSE (for reference)

For completeness we report the aggregate 256-step field RMSE that we argue against using in isolation. M2 data-efficiency (small config, 5 seeds): global $1.560 \pm 0.005 / 1.556 \pm 0.004 / 1.558 \pm 0.002$ at 3/10/30 trajectories vs. local $1.802 \pm 0.247 / 1.709 \pm 0.232 / 1.653 \pm 0.141$. Global’s near-zero variance across data budgets (it does not improve with more data) is diagnostic of blur-coasting, not stability (§5.3).

C Hyperparameters, seeds, and compute

	large config	small config
hidden dim	128	96
depth (blocks)	4	3
window	48	48
training steps	6000	3000
seeds	3	5
batch / kNN	2 / 16	2 / 16
optimizer	AdamW, lr 3×10^{-4} , cosine	
noise std / dissip. weight	0.02 / 0.3	
normalization	scale-only (v)	
hardware	one NVIDIA A100, grad. checkpointing	
approx. params	1.3M	0.6M

Table 4: Configurations, run as a 2×2 dataset $\times$ capacity grid.

Diagnostic: $\epsilon=10^{-2}$ nominal (robust over $[3 \times 10^{-3}, 10^{-1}]$), 30 source nodes on the active front, threshold 2.5 km (robust 2.5–8 km), $n=18$ samples. Front metrics: THR = 0.5 m/s. GNS reference: authors’ released MeshSimulator, 10 message-passing steps, 3M training steps (Liu et al., 2025; Zenodo 17095311).

References

- Uri Alon and Eran Yahav. On the bottleneck of graph neural networks and its practical implications. In *ICLR*, 2021.
- Johannes Brandstetter, Daniel Worrall, and Max Welling. Message passing neural pde solvers. In *ICLR*, 2022.
- Michael M Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv:2104.13478*, 2021.
- Shuhao Cao. Choose a transformer: Fourier or galerkin. In *NeurIPS*, 2021.
- Alexey Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *COLM*, 2024.
- Zongyi Li et al. Fourier neural operator for parametric partial differential equations. In *ICLR*, 2021.
- Zongyi Li et al. Learning chaotic dynamics in dissipative systems. In *NeurIPS*, 2022.
- Phillip Lippe et al. Pde-refiner: Achieving accurate long rollouts with neural pde solvers. In *NeurIPS*, 2023.
- Dunyu Liu et al. Earthquake rupture dynamics from graph neural networks. *Journal of Geophysical Research: Solid Earth*, 2025.
- Tobias Pfaff, Meire Fortunato, Alvaro Sanchez-Gonzalez, and Peter Battaglia. Learning mesh-based simulation with graph networks. In *ICLR*, 2021.
- Nasim Rahaman et al. On the spectral bias of neural networks. In *ICML*, 2019.
- Alvaro Sanchez-Gonzalez, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter Battaglia. Learning to simulate complex physics with graph networks. In *ICML*, 2020.
- Haixu Wu et al. Transolver: A fast transformer solver for pdes on general geometries. In *ICML*, 2024.

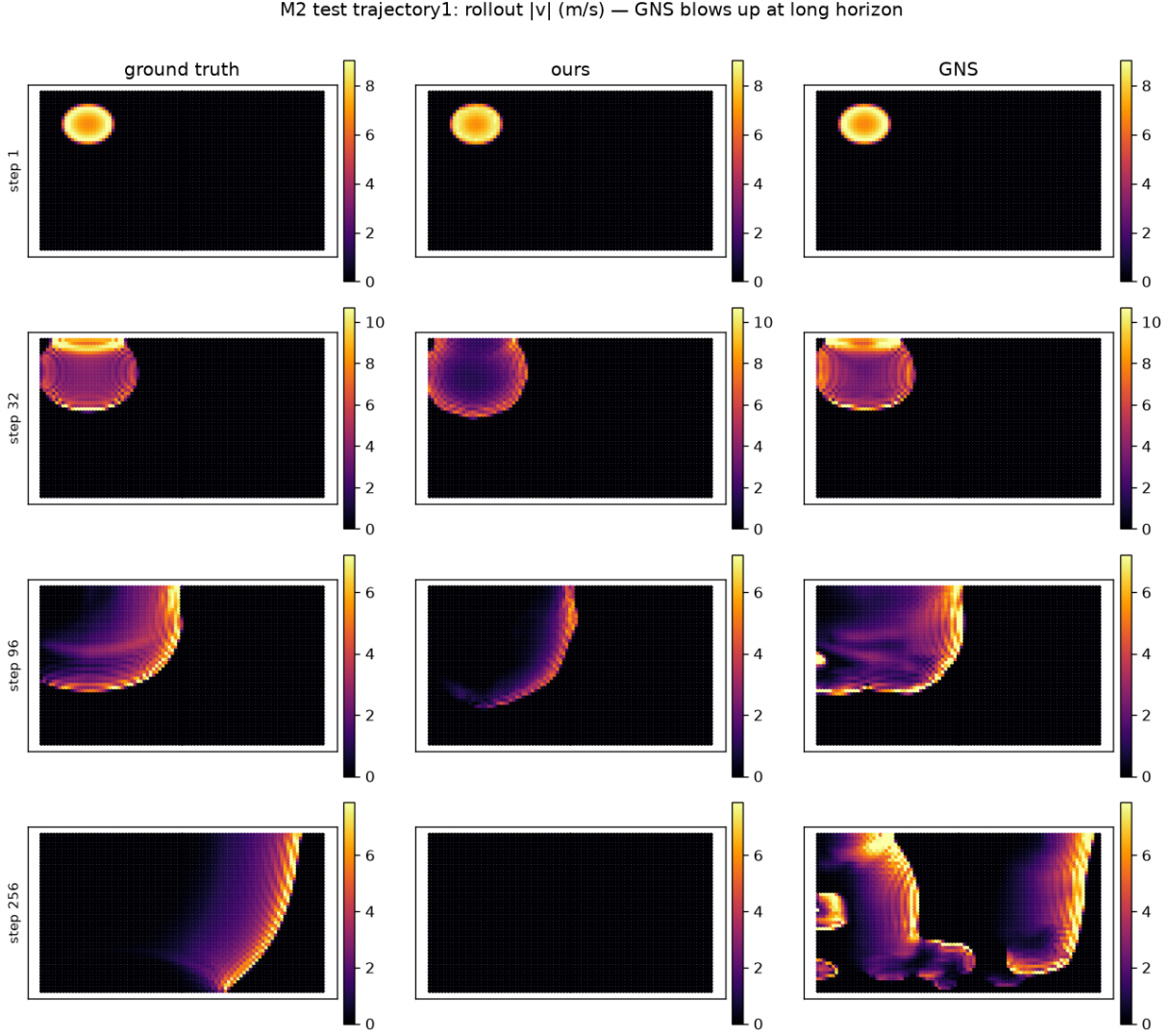


Figure 1: Qualitative rollout comparison of the slip-rate field $|v|$ (m/s) on an M2 test trajectory at steps 1, 32, 96, and 256. The ground truth (left) shows a sharp, propagating rupture front. Local attention (middle) keeps the front sharp and stable but eventually dissipates (dies out) by step 256. GNS (right, shown here for the 80k-step checkpoint) successfully propagates the front but experiences catastrophic high-frequency noise and blow-up at the long horizon (step 256).

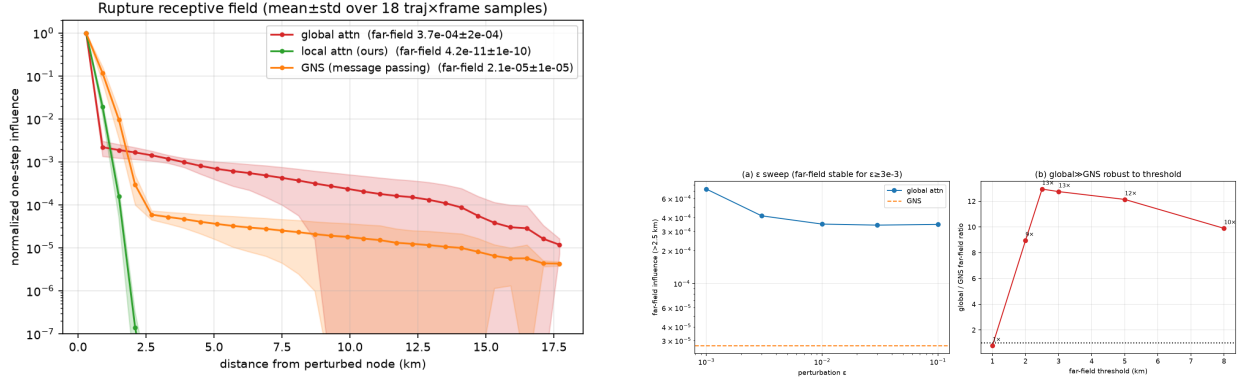


Figure 2: *Left*: effective receptive field on rupture (log scale, mean $\pm$ std, $n=18$), global attention has only a small far-field floor (it is effectively nearly local); local attention is compact. *Right*: the measurement is robust to the perturbation size ϵ and the far-field threshold.

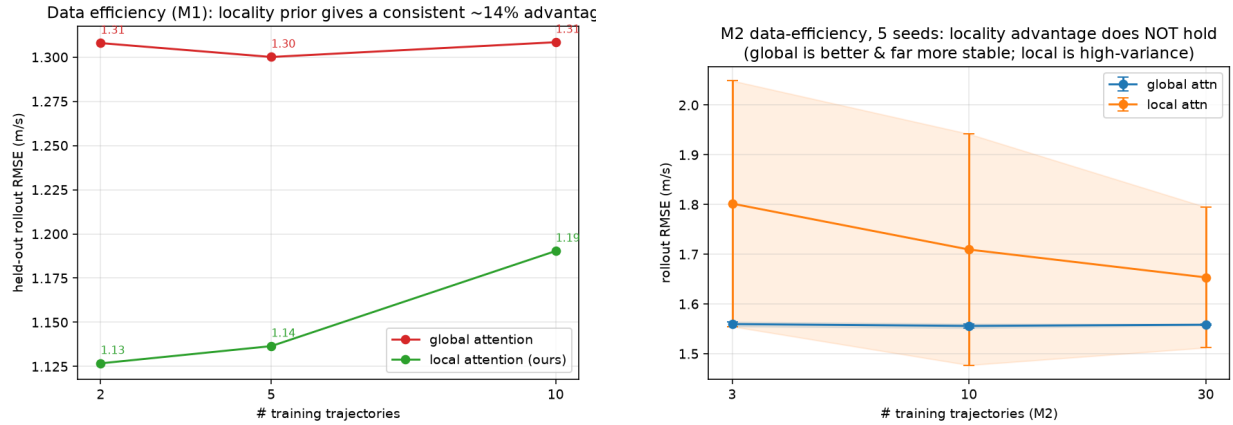


Figure 3: Data efficiency under different settings and metrics. *Left*: On M1 (low diversity), local attention consistently outperforms global attention by $\sim 14\%$ across budgets. *Right*: On M2 (diverse asperities, 5 seeds, small config), under aggregate rollout RMSE, global attention appears to win and remain extremely stable. As detailed in §5.3, this is a spurious artifact of the blur pitfall: the global model completely collapses the propagating front to near-zero (energy ratio 0.02), thereby achieving a flat, low aggregate RMSE on the quiescent background, while local attention attempts to track the front, resulting in a higher and more variable aggregate RMSE due to geometric dilution of front errors.

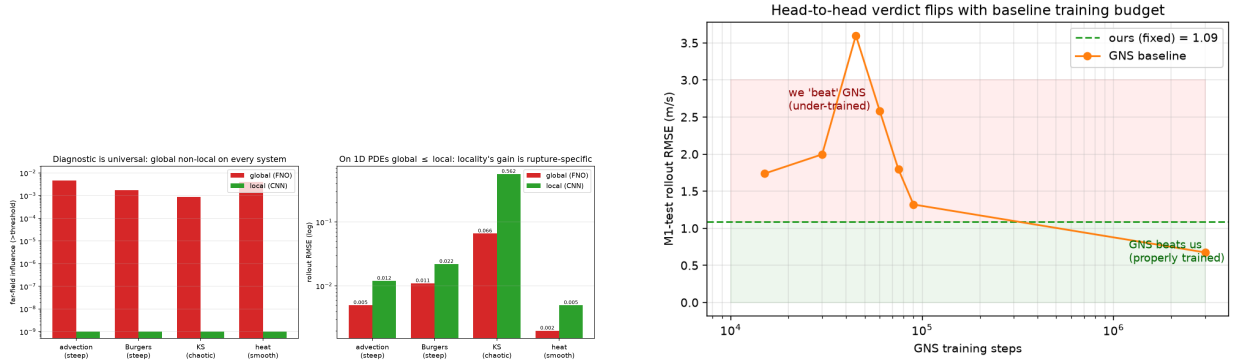


Figure 4: *Left: 1D-PDE sanity check for the diagnostic*, it recovers FNO’s (global) and CNN’s (local) by-design support (left sub-panel), confirming the probe reports known support. (The right sub-panel shows rollout accuracy for reference only; these 1D runs are single-seed and the CNN is unstable on KS, so we do *not* use them as evidence for any accuracy claim.) *Right: evaluation pitfall two*, the head-to-head verdict depends on how long the GNS baseline is trained.